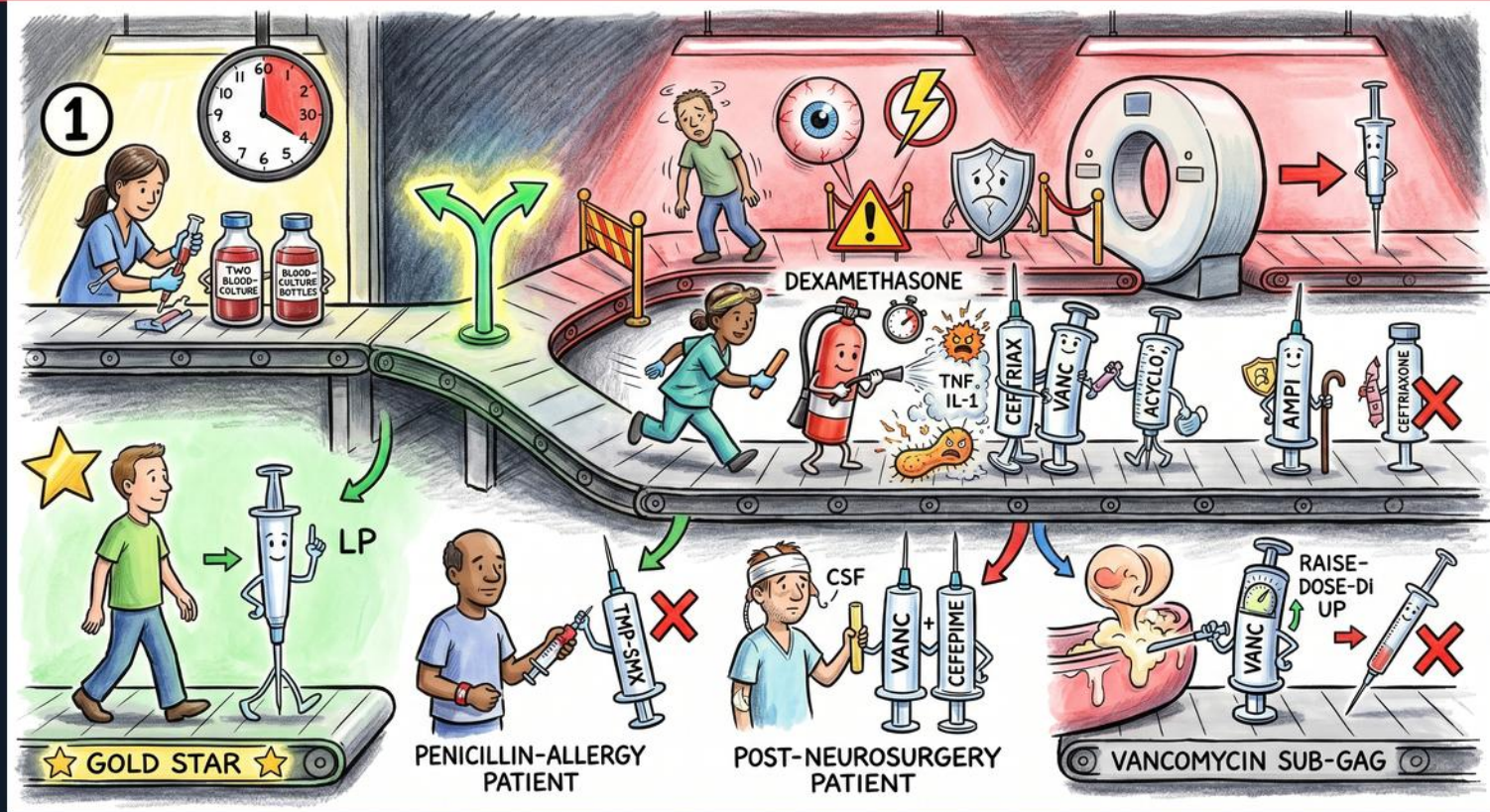


Meningitis — Management Order, Empiric Antibiotics & Dexamethasone





Step 1: blood cultures + clock <60 min

WHAT YOU SEE

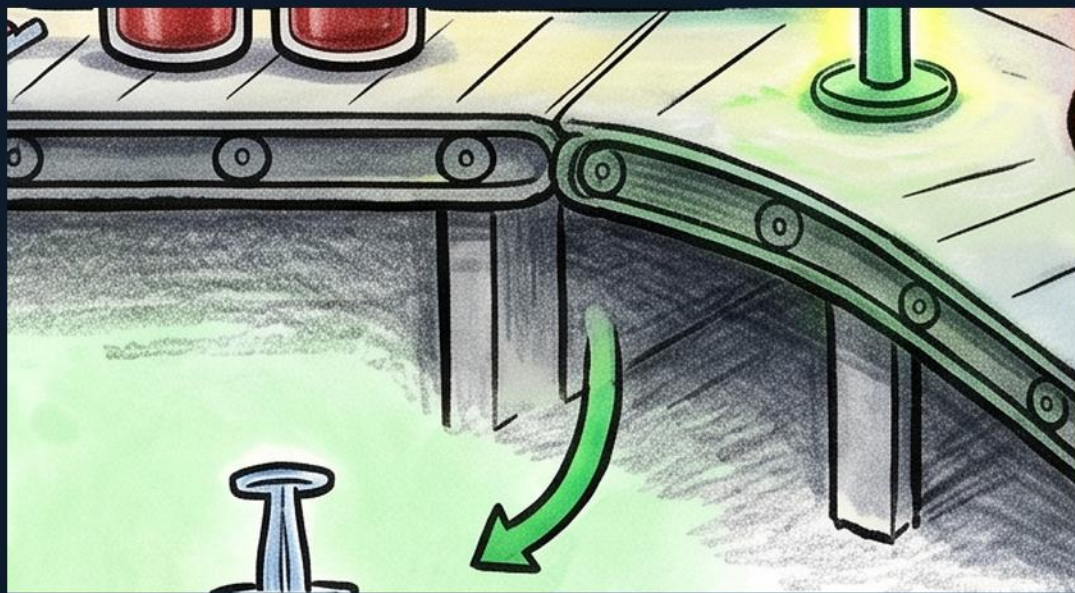
The big numbered '1', wall clock, and two blood-culture vials at the start of the conveyor — draw cultures and start treatment within 60 minutes.

WHAT IT ENCODES

Sequence: draw 2 sets of blood cultures → give dexamethasone with/just before the first antibiotic → give empiric antibiotics, all within 60 minutes (door-to-antibiotic time drives mortality). Blood cultures are positive in ~50-70% and preserve a microbiologic diagnosis even when CSF is sterilized by antibiotics.

THE BOARD TRAP

"wait until CT and LP are done before antibiotics" -> wrong; every hour of delay increases death/disability — give empiric therapy right after blood cultures; CSF yield drops only modestly in the first few hours.



Decision fork: who needs CT first

WHAT YOU SEE

The green forking arrow on the conveyor floor — the branch deciding direct LP versus CT-first.

WHAT IT ENCODES

Obtain CT head BEFORE LP only with an IDSA red flag: focal neurologic deficit, papilledema, new-onset seizure (within 1 week), moderate-severe altered consciousness, immunocompromise (HIV, transplant, immunosuppressants), or known CNS disease/mass. These raise concern for a mass lesion and herniation risk from LP.

THE BOARD TRAP

"CT every suspected meningitis to be safe" -> wrong; universal pre-LP CT delays low-risk patients who can be tapped immediately — reserve CT for the specific IDSA features, and never let it delay antibiotics.



Left path: direct LP (gold star)

WHAT YOU SEE

Gold-star patient walking calmly toward the 'LP' needle on the green belt — no focal signs, proceed straight to lumbar puncture.

WHAT IT ENCODES

Immunocompetent, normal mentation, no focal deficit/papilledema/recent seizure → go **DIRECTLY** to LP without CT, then start dexamethasone + empiric antibiotics. Sending CSF before antibiotics (when feasible without delay) maximizes Gram stain and culture yield.

THE BOARD TRAP

"normal neuro exam means you can delay antibiotics for the LP" → wrong; even in the direct-LP path, if obtaining CSF is held up, give antibiotics immediately after blood cultures.



Right path: CT-first red flags

WHAT YOU SEE

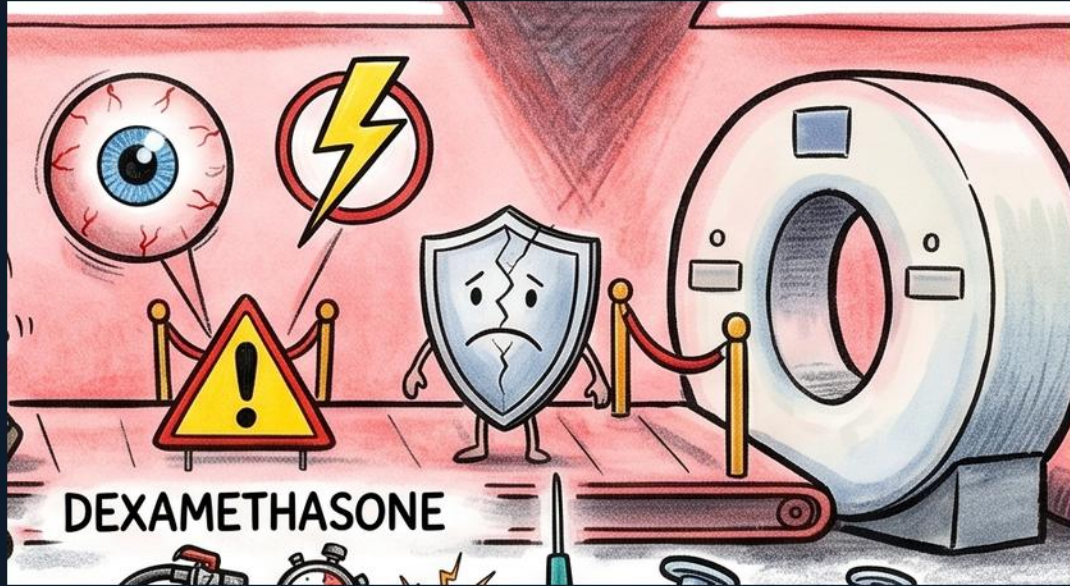
Red-lit alarm zone with eye (papilledema), lightning bolt (seizure), and the CT scanner — the concerning features that mandate imaging before LP.

WHAT IT ENCODES

Concerning features → blood cultures + dexamethasone + empiric antibiotics FIRST, THEN CT to exclude a mass/herniation risk, THEN LP if CT is clear. The antibiotics-before-CT order is the high-yield point: imaging must never push back the first dose.

THE BOARD TRAP

"CT → LP → then antibiotics" → wrong; correct order when CT is indicated is blood cultures + steroids + antibiotics → CT → LP. Waiting for imaging before treating is the classic error.



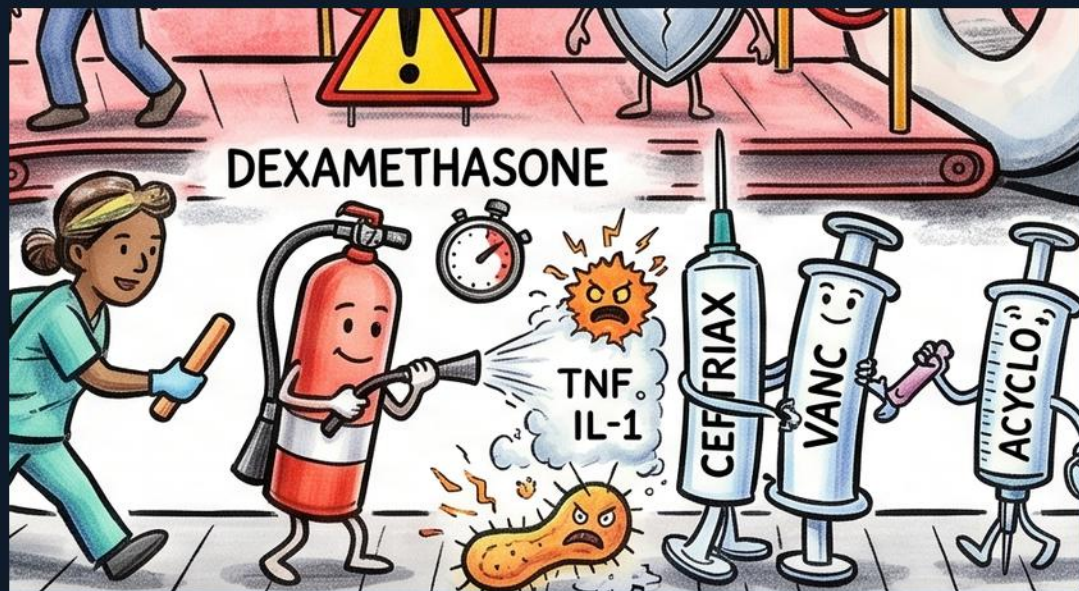
CT scanner

WHAT YOU SEE

The donut CT scanner on the upper belt — the imaging step reserved for red-flag patients only.

WHAT IT ENCODES

CT head before LP excludes a space-occupying lesion / herniation risk in red-flag patients. It is NOT required for low-risk patients and must never delay the first antibiotic dose. Most herniation in meningitis is from diffuse cerebral edema, so even a normal CT does not fully eliminate risk.



Empiric: ceftriaxone + vancomycin + acyclovir

WHAT YOU SEE

The line of labeled syringes — CEF/TRIAX, VANC, ACYCLO — the standard immunocompetent-adult empiric trio.

WHAT IT ENCODES

Standard empiric regimen for immunocompetent adults: ceftriaxone 2g IV q12h (*S. pneumoniae*, *N. meningitidis*, *H. influenzae*) PLUS vancomycin 15-20 mg/kg IV q8-12h (ceftriaxone-resistant pneumococcus). Add acyclovir 10 mg/kg IV q8h if HSV encephalitis is on the differential (altered mentation, temporal-lobe signs, seizures).

THE BOARD TRAP

"ceftriaxone monotherapy" -> wrong; up to ~25% of pneumococci have reduced cephalosporin susceptibility, so vancomycin is mandatory empirically until susceptibilities return.



Add ampicillin >50 / immunocompromised

WHAT YOU SEE

The AMPI syringe at the end of the row — added *Listeria* coverage for the older or immunocompromised host.

WHAT IT ENCODES

Add AMPICILLIN 2g IV q4h for *Listeria* in age >50, neonates, pregnancy, alcoholism, and immunocompromise (steroids, transplant, HIV, malignancy). Cephalosporins have NO *Listeria* activity, so ampicillin (or penicillin G) is the dedicated agent for this gram-positive rod.

THE BOARD TRAP

"ceftriaxone + vancomycin alone for a 65-year-old" -> wrong; cephalosporins miss *Listeria* — omitting ampicillin in a *Listeria*-risk host leaves a fatal gap.



PCN allergy: TMP-SMX (red X)

WHAT YOU SEE

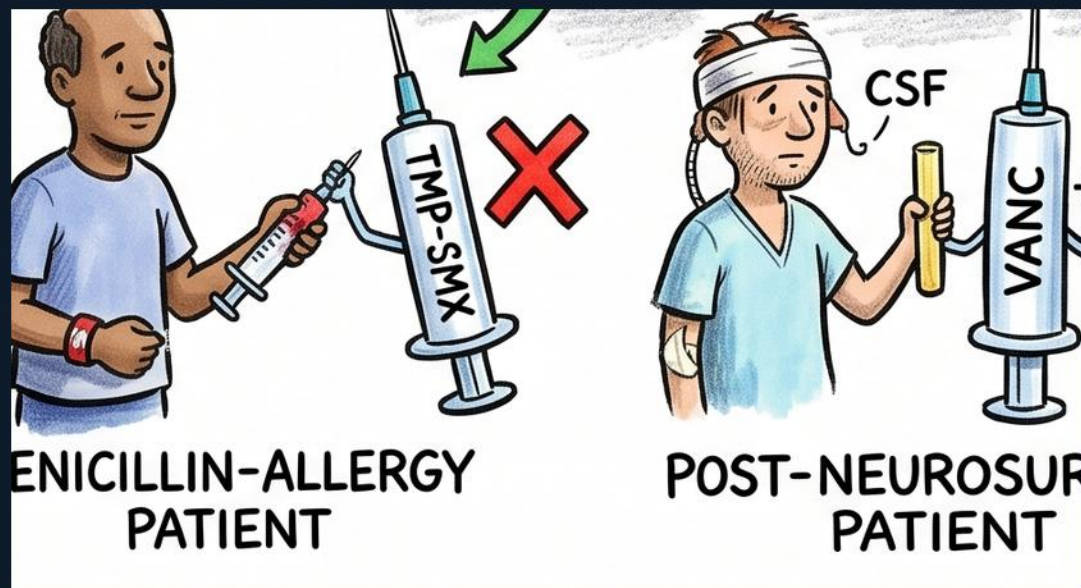
Penicillin-allergy patient with a red-X over the TMP-SMX syringe — the Listeria swap for true penicillin allergy.

WHAT IT ENCODES

For Listeria coverage in true penicillin allergy, use TMP-SMX (Bactrim) IV; meropenem is an alternative covering Listeria plus gram-negatives (low cross-reactivity except in severe/anaphylactic histories).

THE BOARD TRAP

"swap ampicillin for cefepime in a PCN-allergic Listeria-risk patient" -> wrong; NO cephalosporin covers Listeria — the correct swap is TMP-SMX or meropenem, not another beta-lactam in the cephalosporin class.



Nosocomial / post-neurosurgery: vanc + cefepime

WHAT YOU SEE

Bandaged post-neurosurgery patient with VANC and CEFEPIME syringes — healthcare-associated coverage including *Pseudomonas*.

WHAT IT ENCODES

Healthcare-associated meningitis (post-neurosurgery, CSF shunt/drain, penetrating trauma, neutropenic) needs vancomycin (MRSA, coag-negative staph) PLUS an antipseudomonal beta-lactam: cefepime, ceftazidime, or meropenem. These hosts harbor staphylococci and *Pseudomonas* not covered by ceftriaxone.

THE BOARD TRAP

"community ceftriaxone-based therapy after neurosurgery/shunt" -> wrong; ceftriaxone misses *Pseudomonas* and under-covers staph — the healthcare setting mandates antipseudomonal + MRSA coverage.



Dexamethasone BEFORE antibiotics

WHAT YOU SEE

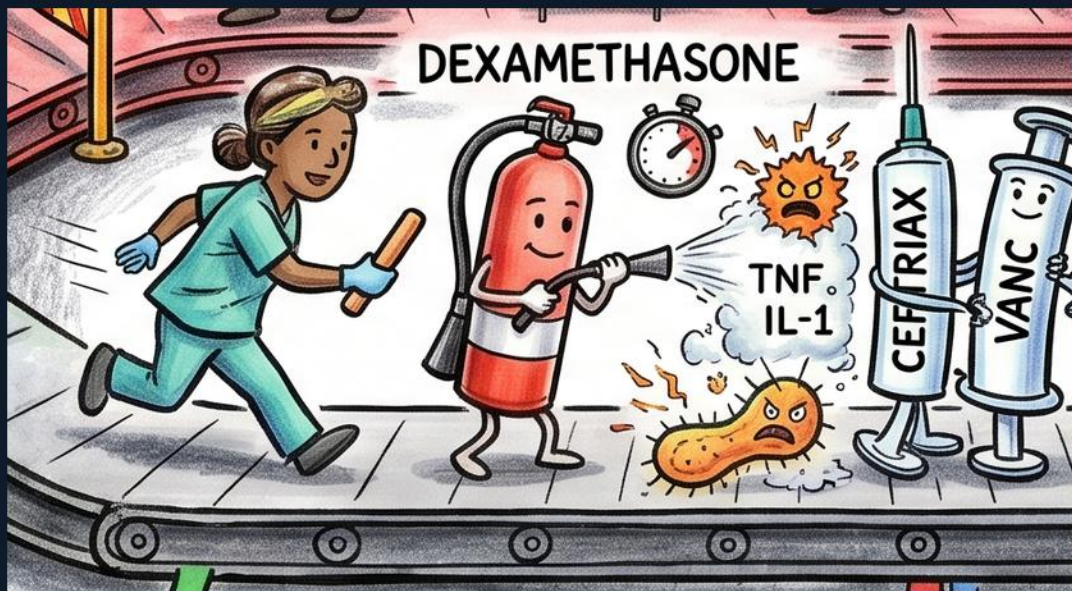
The bomb-fuse 'DEXAMETHASONE' character lit on the belt — give steroid 15-20 min before/with the first antibiotic.

WHAT IT ENCODES

Give dexamethasone 10mg IV 15-20 minutes BEFORE, or concurrently with, the first antibiotic in suspected bacterial meningitis. Antibiotics lyse bacteria releasing cell-wall fragments that trigger a cytokine surge; pre-dosing blunts this storm and reduces death and sequelae (notably pneumococcal).

THE BOARD TRAP

"give dexamethasone after antibiotics are already running" -> wrong; late steroid offers little benefit once bacteriolysis/cytokines are underway — but also do NOT delay antibiotics for steroid; give together if steroid isn't ready.



Mechanism: TNF-a / IL-1B from lysis

WHAT YOU SEE

The bursting bacterium spilling 'TNF-IL-1' near the macrophages — cell-wall release driving the cytokine cascade.

WHAT IT ENCODES

Mechanism: antibiotics kill bacteria, releasing cell-wall components (pneumococcal peptidoglycan, LPS) that activate macrophages/microglia to release TNF-alpha and IL-1beta, driving meningeal inflammation, cerebral edema, vasculitis, and raised ICP. Dexamethasone suppresses this host response, limiting secondary CNS injury.

THE BOARD TRAP

"the steroid treats the infection" -> wrong; dexamethasone is a purely anti-inflammatory adjunct — it does nothing to the organism and must accompany, never replace, antibiotics.



Dexa benefit greatest in pneumococcus

WHAT YOU SEE

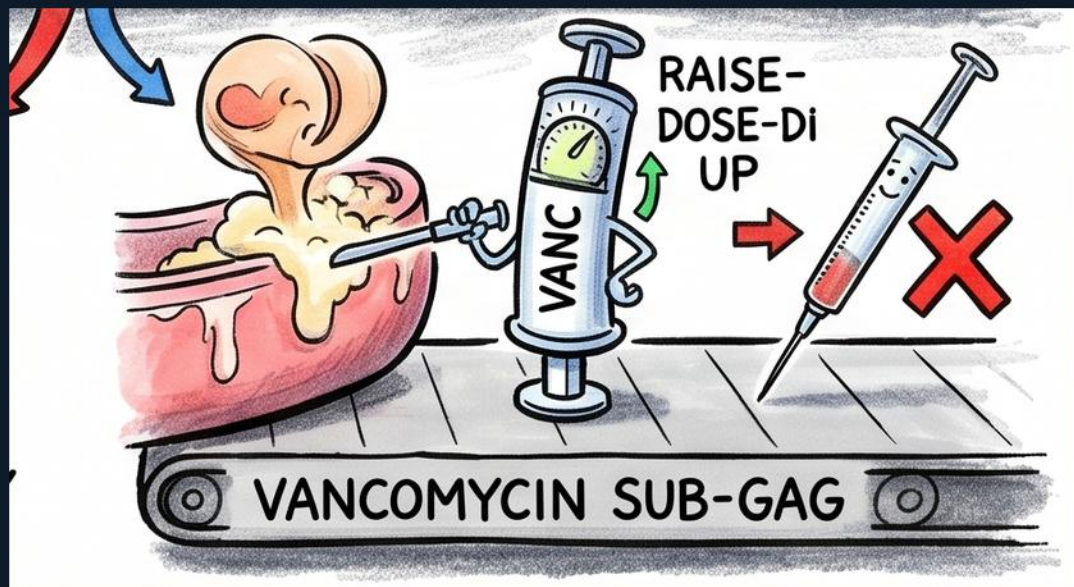
The shield character guarding the belt — dexamethasone's proven benefit, strongest for *S. pneumoniae*.

WHAT IT ENCODES

Dexamethasone's proven mortality benefit is greatest in PNEUMOCOCCAL meningitis in adults; in children it reduces sensorineural HEARING LOSS in Hib meningitis. Dose: 10mg IV q6h x4 days (peds 0.15 mg/kg q6h x4d). Continue only if cultures confirm pneumococcus (adults) or Hib (children); stop otherwise. Benefit is established mainly in high-income settings.

THE BOARD TRAP

"finish the full 4-day steroid course regardless of organism" -> wrong; for meningococcal, Listeria, gram-negative, or viral disease benefit is unproven — STOP dexamethasone once those are confirmed.



Vancomycin sub-gag: raise the dose

WHAT YOU SEE

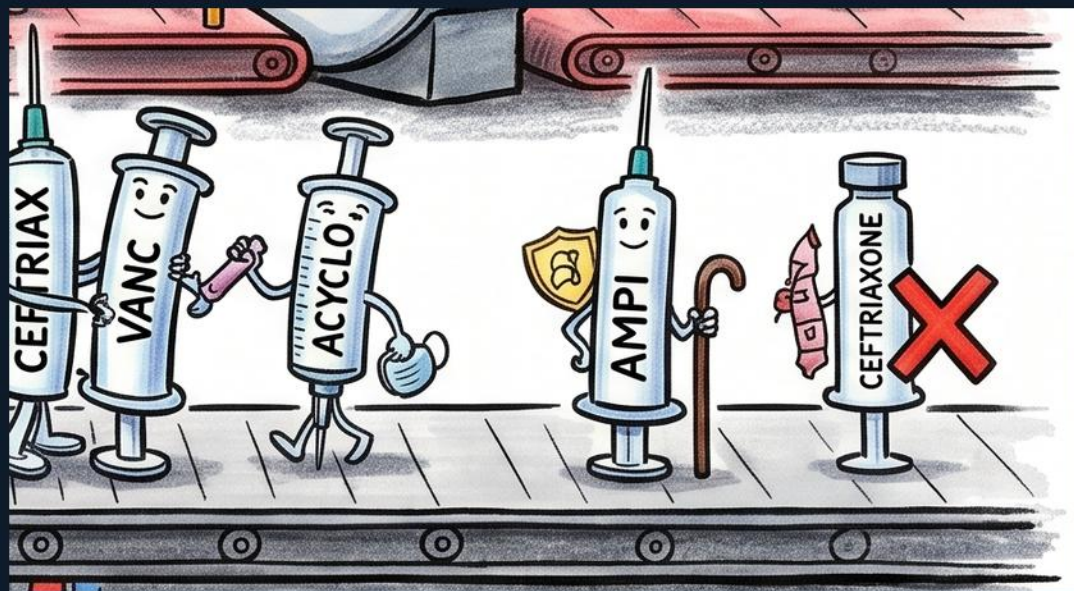
Patient in the bath with VANC and a red-X over 'raise-dose' — dexamethasone cuts vancomycin CSF penetration, so increase the dose.

WHAT IT ENCODES

Dexamethasone reduces meningeal inflammation and BBB permeability, lowering vancomycin CSF penetration. When giving dexamethasone WITH vancomycin for resistant pneumococcus, use higher dosing (trough 15-20, ~45-60 mg/kg/day) and consider adding rifampin if response is poor.

THE BOARD TRAP

"give vancomycin intrathecally to overcome poor penetration" —> wrong; the fix is to RAISE IV vancomycin and follow troughs (or add rifampin) — routine intrathecal vancomycin isn't standard for community-acquired meningitis and is toxic.



Acyclovir vs ganciclovir

WHAT YOU SEE

The lone CEFTRIAXONE/antiviral syringe with a red X at the far right — the HSV-vs-CMV antiviral distinction.

WHAT IT ENCODES

Add acyclovir 10 mg/kg IV q8h empirically when HSV/VZV encephalitis is suspected (fever + altered mentation, focal/temporal-lobe signs, seizures, CSF lymphocytosis with RBCs). Acyclovir covers HSV-1/2 and VZV; CMV requires ganciclovir/foscarnet. Hydrate to prevent acyclovir crystalline nephropathy.

THE BOARD TRAP

"acyclovir for CMV encephalitis" -> wrong; acyclovir has essentially no CMV activity (CMV lacks thymidine kinase) — use ganciclovir/foscarnet for CMV, acyclovir for HSV/VZV.

Summary — every symbol

1. Step 1: blood cultures + clock <60 min

Sequence: draw 2 sets of blood cultures → give dexamethasone with/just before the first antibiotic → give empiric antibiotics, all within 60 minutes (door-to-antibiotic time drives mortality)

3. Left path: direct LP (gold star)

Immunocompetent, normal mentation, no focal deficit/papilledema/recent seizure → go DIRECTLY to LP without CT, then start dexamethasone + empiric antibiotics

5. CT scanner

CT head before LP excludes a space-occupying lesion / herniation risk in red-flag patients

7. Add ampicillin >50 / immunocompromised

Add AMPICILLIN 2g IV q4h for Listeria in age >50, neonates, pregnancy, alcoholism, and immunocompromise (steroids, transplant, HIV, malignancy)

9. Nosocomial / post-neurosurgery: vanc + cefepime

Healthcare-associated meningitis (post-neurosurgery, CSF shunt/drain, penetrating trauma, neutropenic) needs vancomycin (MRSA, coag-negative staph) PLUS an antipseudomonal beta-lactam: cefepime, ceftazidime, or meropenem

11. Mechanism: TNF- α / IL-1 β from lysis

Mechanism: antibiotics kill bacteria, releasing cell-wall components (pneumococcal peptidoglycan, LPS) that activate macrophages/microglia to release TNF-alpha and IL-1beta, driving meningeal inflammation, cerebral edema, vasculitis, and raised ICP

13. Vancomycin sub-gag: raise the dose

Dexamethasone reduces meningeal inflammation and BBB permeability, lowering vancomycin CSF penetration

2. Decision fork: who needs CT first

Obtain CT head BEFORE LP only with an IDSA red flag: focal neurologic deficit, papilledema, new-onset seizure (within 1 week), moderate-severe altered consciousness, immunocompromise (HIV, transplant, immunosuppressants), or known CNS disease/mass

4. Right path: CT-first red flags

Concerning features → blood cultures + dexamethasone + empiric antibiotics FIRST, THEN CT to exclude a mass/herniation risk, THEN LP if CT is clear

6. Empiric: ceftriaxone + vancomycin + acyclovir

Standard empiric regimen for immunocompetent adults: ceftriaxone 2g IV q12h (S)

8. PCN allergy: TMP-SMX (red X)

For Listeria coverage in true penicillin allergy, use TMP-SMX (Bactrim) IV; meropenem is an alternative covering Listeria plus gram-negatives (low cross-reactivity except in severe/anaphylactic histories).

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12. Dexamethasone benefit greatest in pneumococcus

Dexamethasone's proven mortality benefit is greatest in PNEUMOCOCCAL meningitis in adults; in children it reduces sensorineural HEARING LOSS in Hib meningitis

14. Acyclovir vs ganciclovir

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